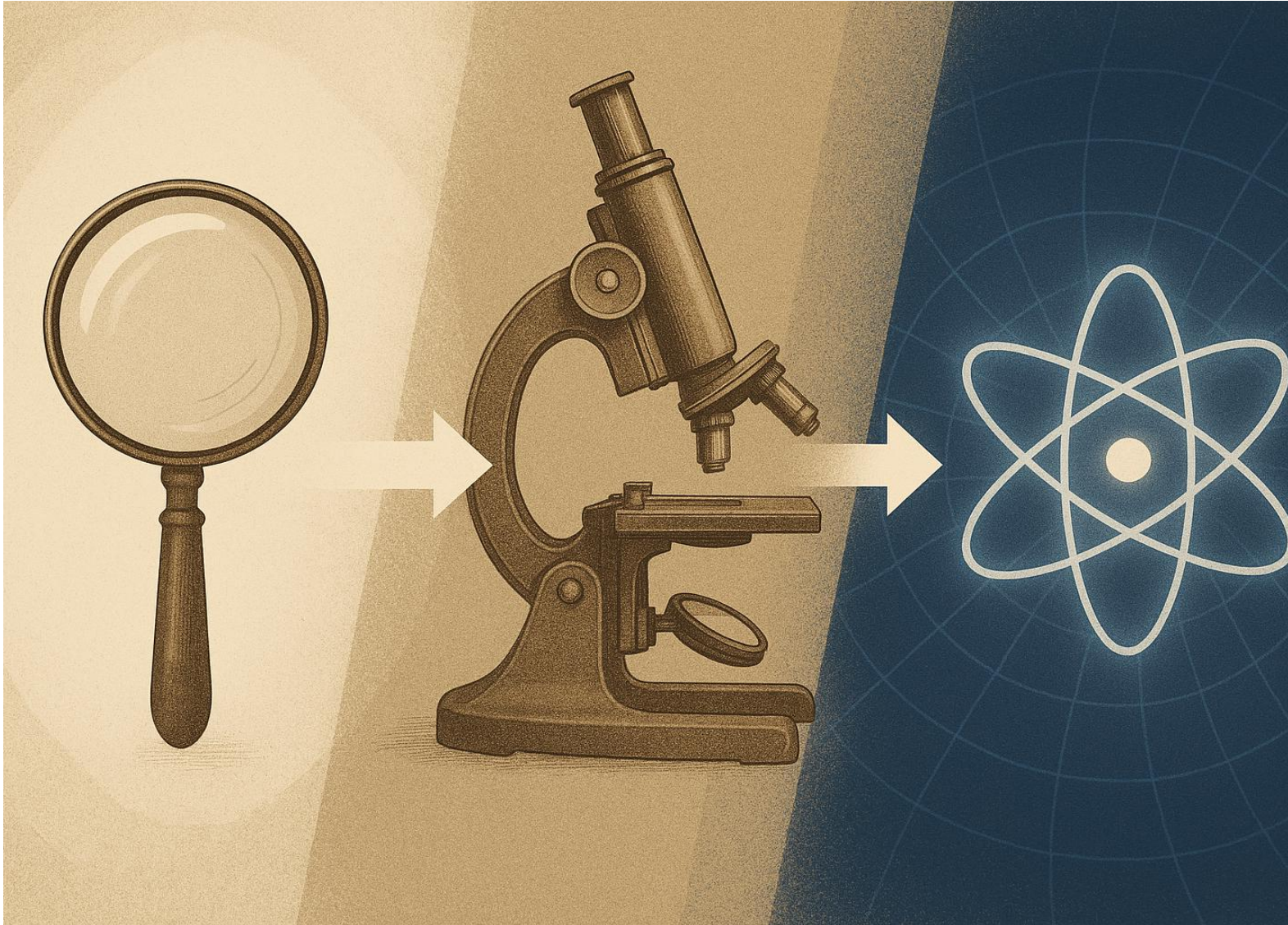

From Firewalls to Fairness

*Securing the Future with Ethics and
Threat Modeling*

Abhilasha Bhargav-Spantzel, Ph.D.
Microsoft Partner Architect





Evolution of Observation

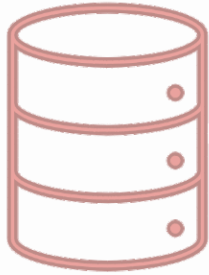
- "My work...was not pursued in order to gain the praise I now enjoy, but chiefly from a craving after knowledge."
Antonie van Leeuwenhoek

Core Security Pillars



Identity Protection

- Ensuring secure and verified access control to prevent unauthorized access and protect sensitive information.



Data Protection

- Protecting sensitive and critical information from unauthorized access, leaks, and breaches to maintain data integrity.



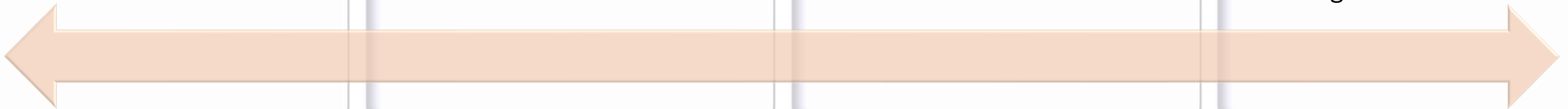
Threat Intelligence

- Gathering and analyzing threat information to stay ahead of potential security risks and enhance defense strategies.



Resilience and Patching

- Maintaining system robustness and promptly addressing vulnerabilities through regular updates, incident response and patch management.



Effective and Efficient Threat Modeling

- 1** Know your product
 - Functionality, Security & Privacy
- 2** Know your assets
 - Scope your critical assets
- 3** Know your adversaries and risks
 - Sophisticated adversaries
 - Insider Risk
- 4** Build Mitigations - Security by Design
 - Built-in security features and mitigations



Asking the many questions



Technology and Humanity

Interplay of Cybersecurity with Ethics

AI is transforming every industry

Proactive and holistic threat modeling

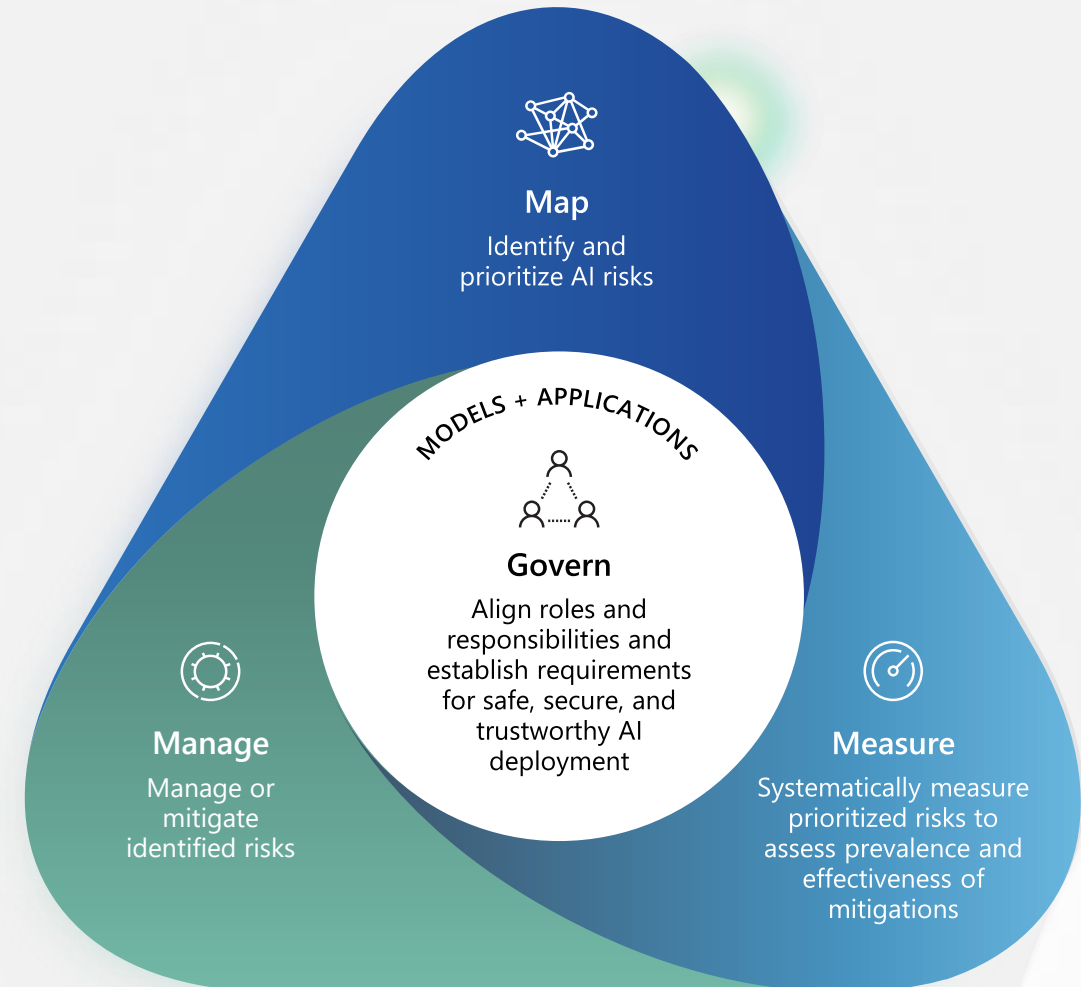
- Fundamental security principles
- Ethical frameworks

Ensuring Safety and Fairness

- Advancing technology with fairness
- Protecting systems and society values



Building generative AI systems responsibly



NIST AI Risk Management Framework

Responsible AI Principles

Transparency

- Ensuring AI systems are understandable

Accountability

- Holding entities responsible for AI actions

Fairness

- Preventing bias and discrimination

Inclusiveness

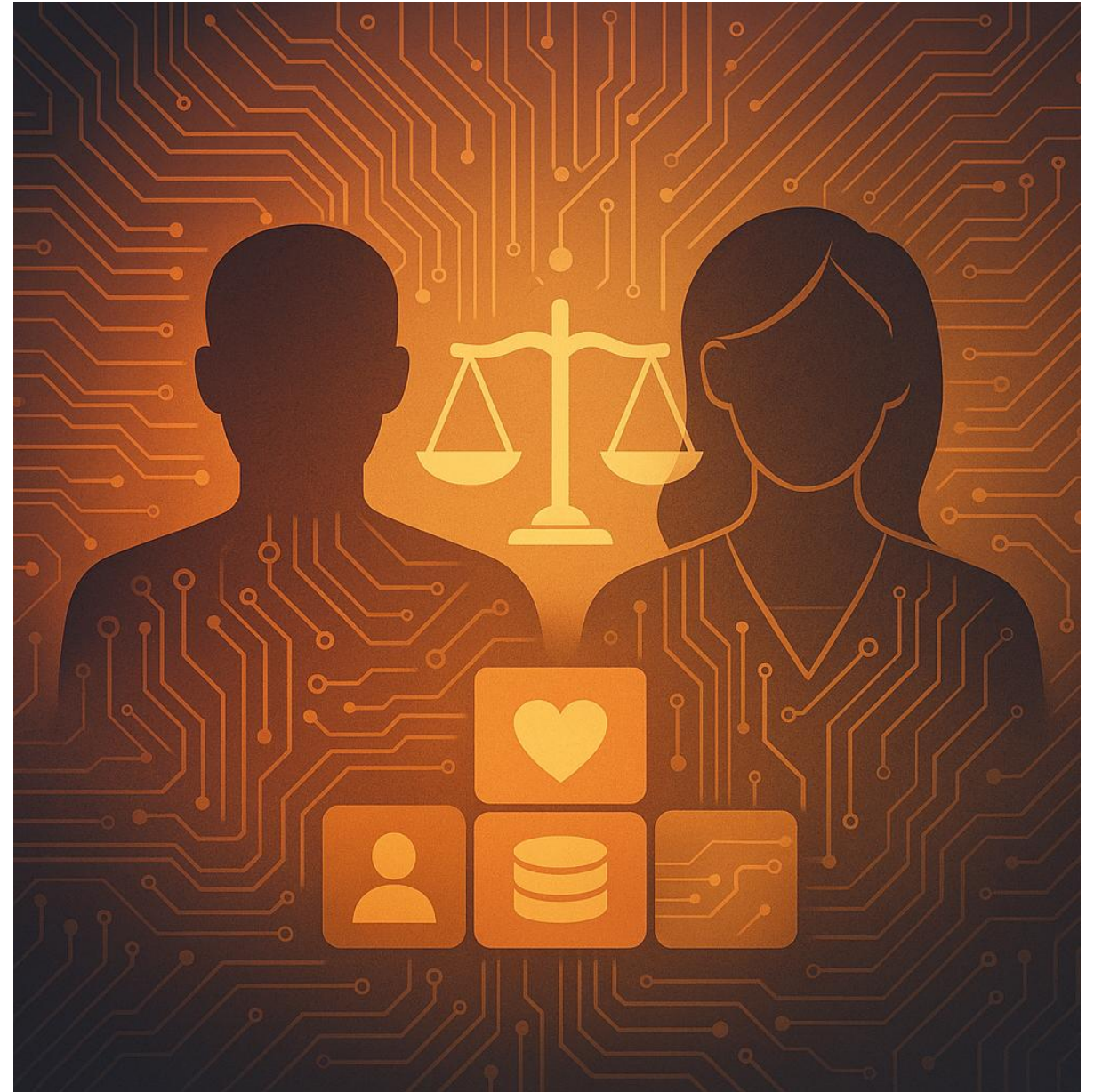
- Considering diverse perspectives

Reliability

- Ensuring consistent performance

Safety

Protecting Human Values and Emotions



Example: XPIA Attacks

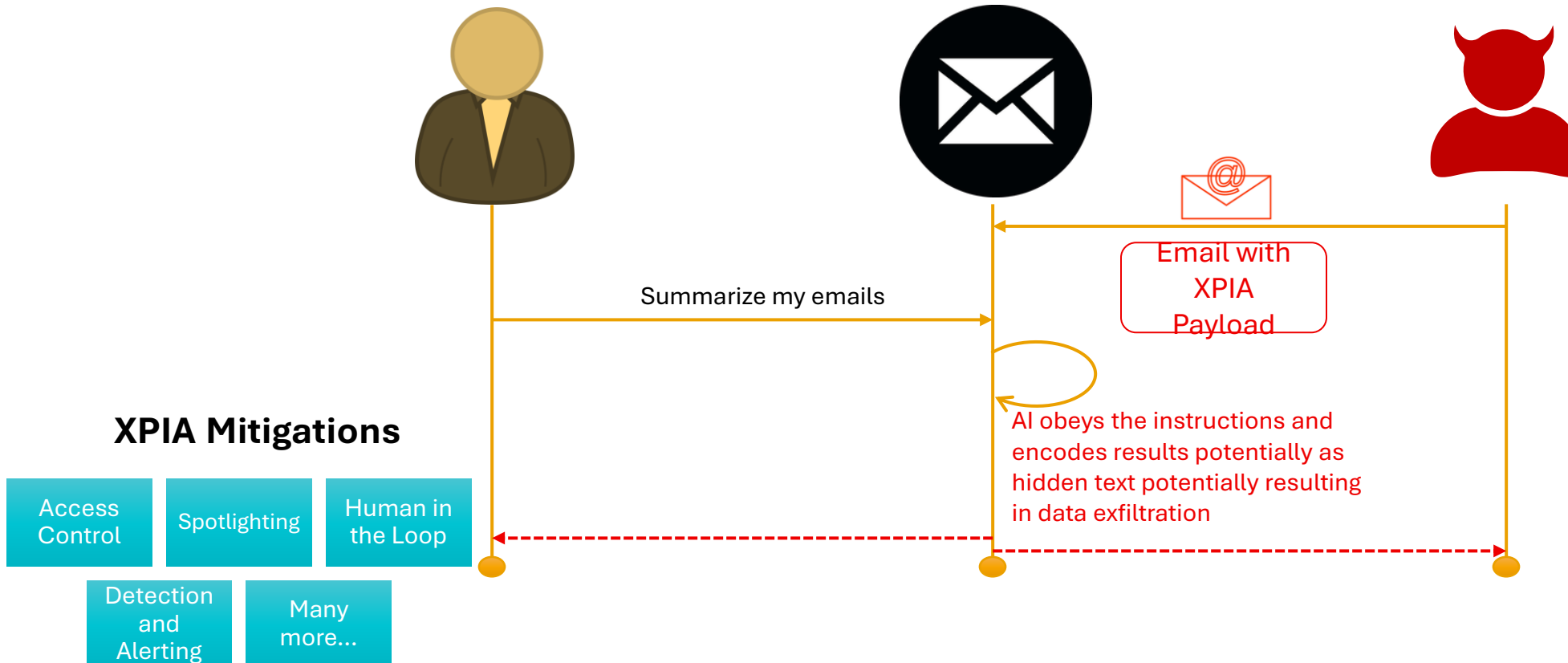
XPIA Attack Steps

Malicious
Prompt Creation

Data Ingestion

Execution

Attacks exploit
LLMs



Strong and Inclusive Adaptive Authentication

Identity as the First Line of Defense

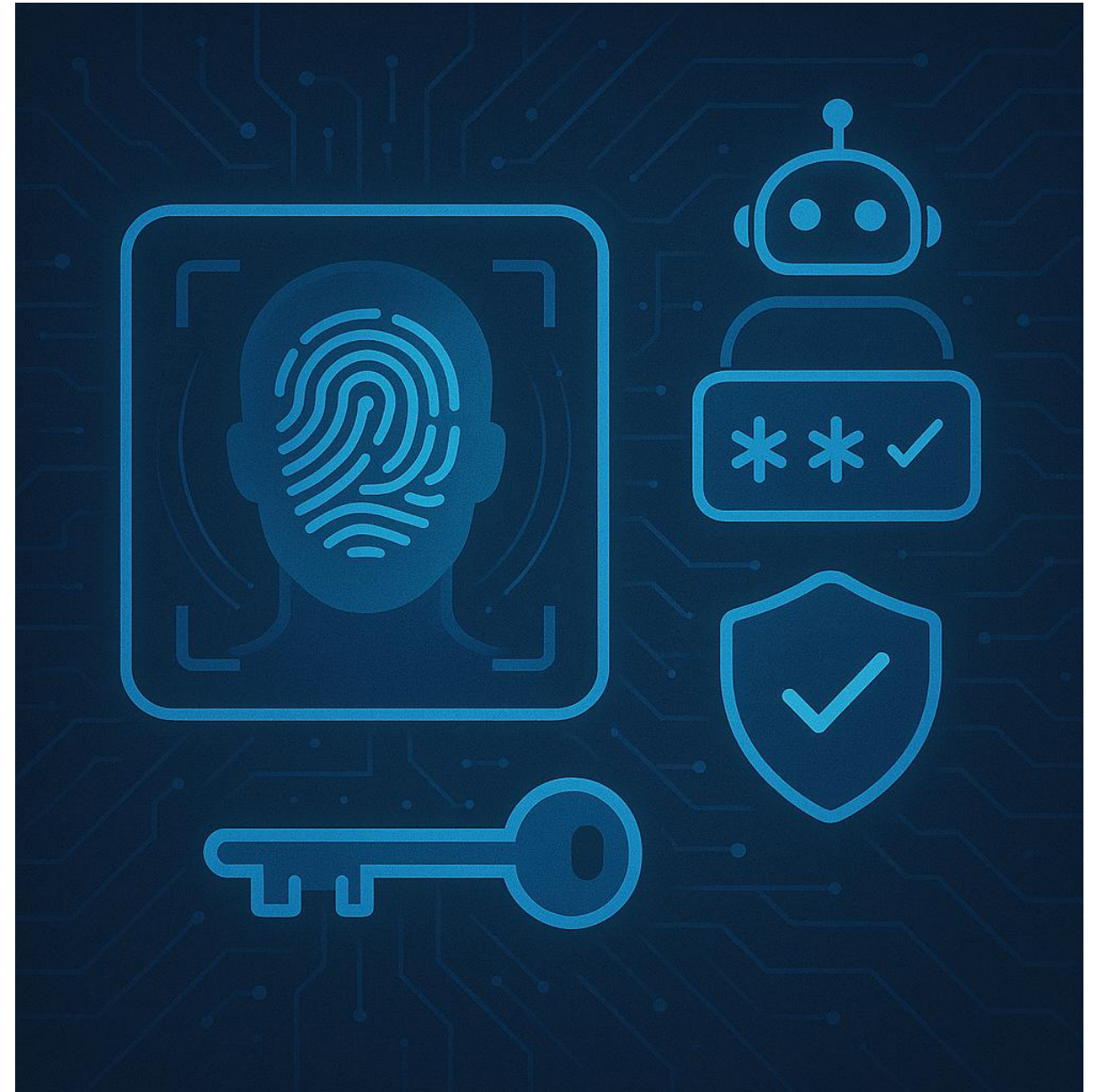
- Importance of securing identity flows
- Techniques like MFA and Zero Trust

AI for Adaptive Authentication

- Enhancing security measures
- Guarding access points with intelligence

Expanding Security Measures for Safety

- Precision in guarding access points
- Intelligent security flows based on age, values



Data Classification and Encryption



Data Classification

- Organizing data based on sensitivity
- Ensuring proper handling of classified data

Data Encryption

- Protecting data through encryption techniques
- Maintaining confidentiality and integrity

Secure APIs

- Implementing secure communication protocols
- Preventing unauthorized access

Monitoring

- Continuous surveillance of data activities
- Identifying and responding to security threats

Attack Surface Reduction & Threat Hunting Strategies

Evolution of Threat Landscape

- Sophisticated adversaries
- Includes AI-generated attacks and coordinated supply chain threats

Understanding Attacker's Capabilities

- Crucial for effective defense

Reducing Attack Surface

- Minimizes potential entry points for attackers

AI-Driven Threat Hunting

- Proactively identifies and mitigates threats
- Helps in staying ahead of adversaries



Maintaining Patch Velocity and Supply Chain Security

Importance of Recovery

No system is invulnerable

Recovery is as important as prevention



Rapid Detection and Rollback Strategies

Powered by AI for anomaly detection

Ensures quick response

Maintains resilience in unforeseen incidents

Preventing Manipulative or Harmful Uses of Technology

Ethical Soundness of Intelligent Systems

Importance of responsible AI

Safeguarding data and assets

Protecting human dignity and values



Proactive Measures Against Abuses

Preventing manipulation and brainwashing

Ensuring positive use of technology

Addressing Advanced Attacks like Cross Prompt Injection (XPIA)

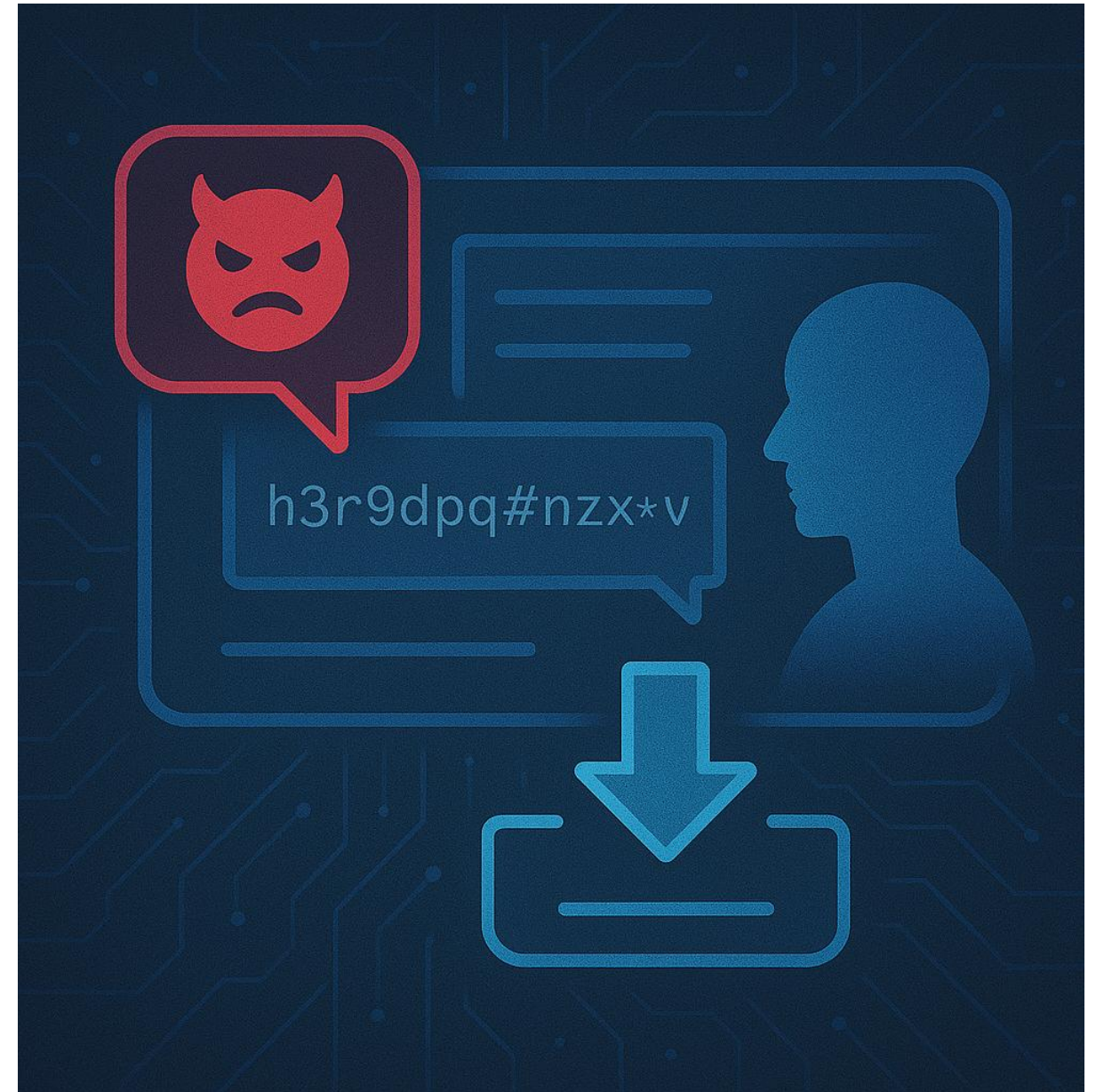
Strong Authentication

Data Flows Analysis

Monitoring for Anomalies

Remediation and Feedback Loop

Preserving Trust with RAI controls



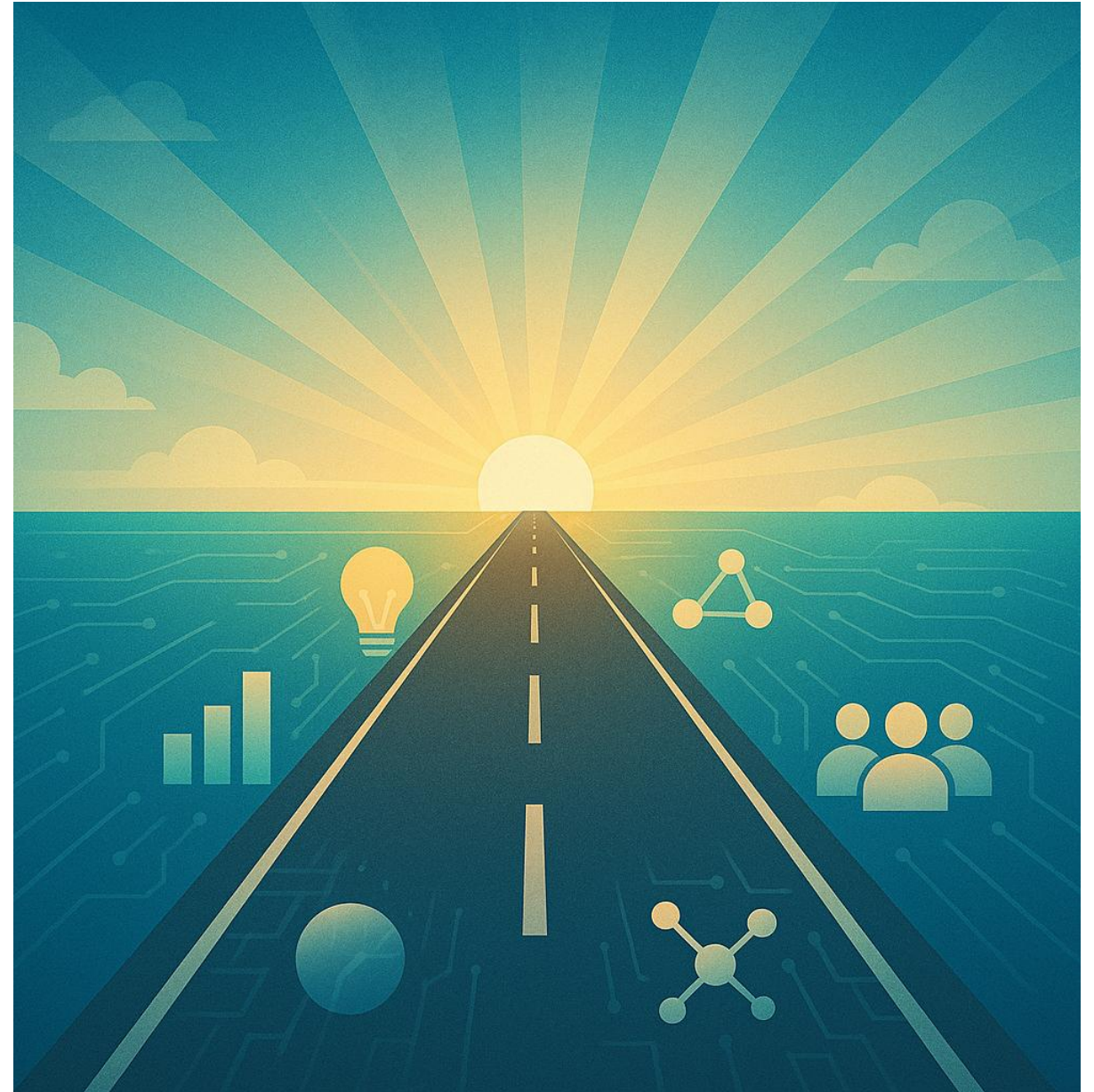
Call to Proactive Collaboration

Ongoing Journey from Firewalls to Fairness

- Combining rigorous threat modeling with ethical frameworks
- Building systems that are secure and uphold collective values

Anticipating Threats Together

- Shaping a future where innovation and responsibility coexist





Thank you!